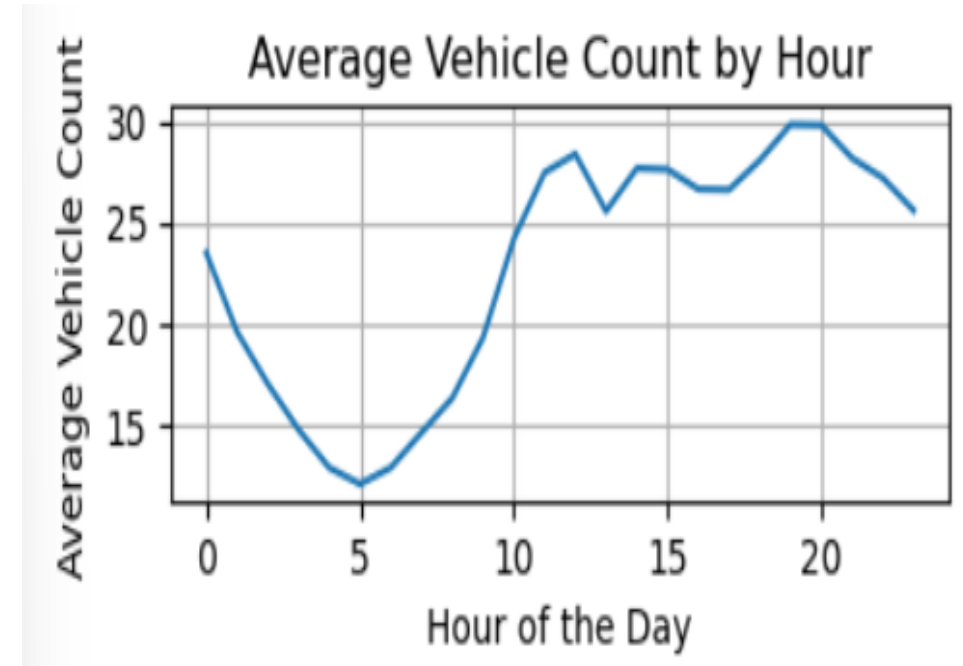


Comprehensive Report on Peak Hour Traffic Analysis

Sheetal Rawat

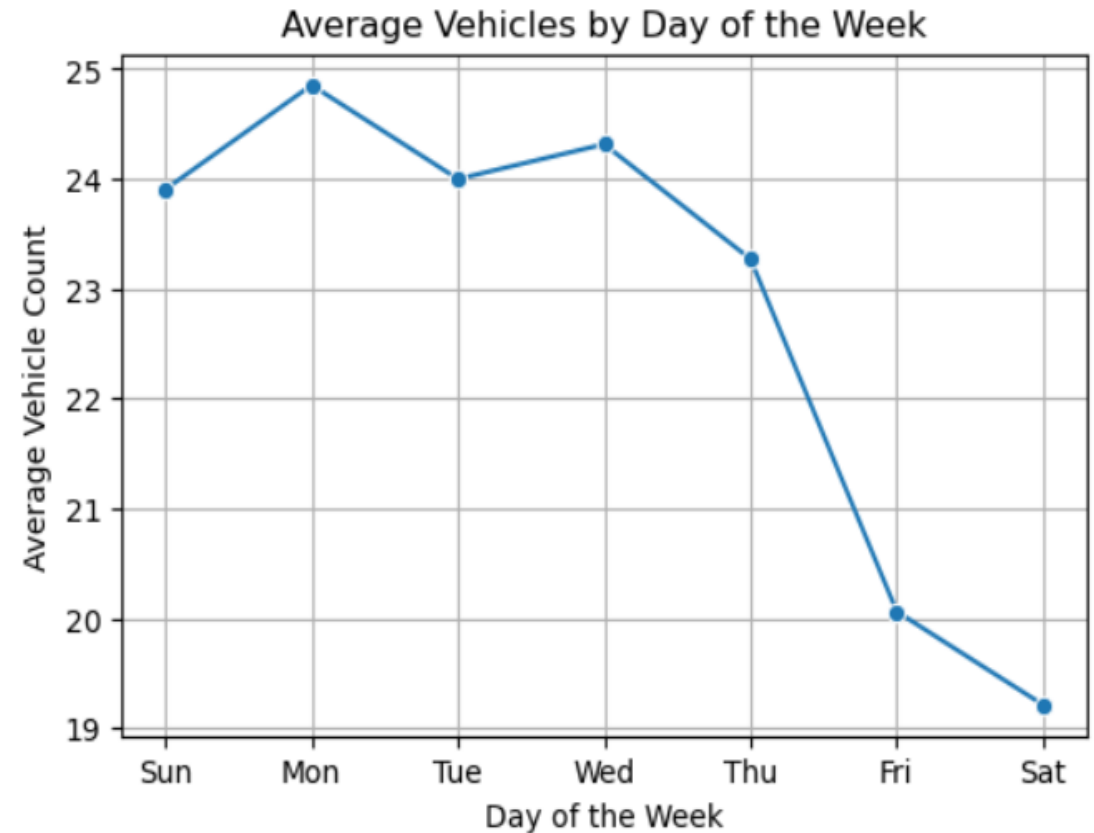
Peak Hour Analysis

- Morning Peak:** The average vehicle count starts high at the beginning of the day and then steadily decreases until around 5 am, indicating the morning rush hour.
- Afternoon Peak:** The count increases again, peaking around 12 pm, reflecting a midday peak, possibly due to people returning to work.
- Second Peak:** Another peak occurs around 8 pm, indicating an evening rush hour.
- Low Traffic:** The average vehicle count remains consistently low between 5 am and 8 am.



Days of the Week Analysis

- Trend:** The graph shows a decreasing trend in the average number of vehicles from Sunday to Saturday.
- Highest Traffic:** Monday has the highest average vehicle count, exceeding 25.
- Lowest Traffic:** Saturday has the lowest average vehicle count, at around 19.
- Weekends vs. Weekdays:** The graph suggests higher traffic on weekdays compared to weekends, with a notable drop in traffic on Fridays and Saturdays.



Months trends

Trend: The graph shows a general trend of increasing vehicle count from January to June, followed by a decrease from July to December.

Peak: The highest average vehicle count is observed in June, followed by September and October.

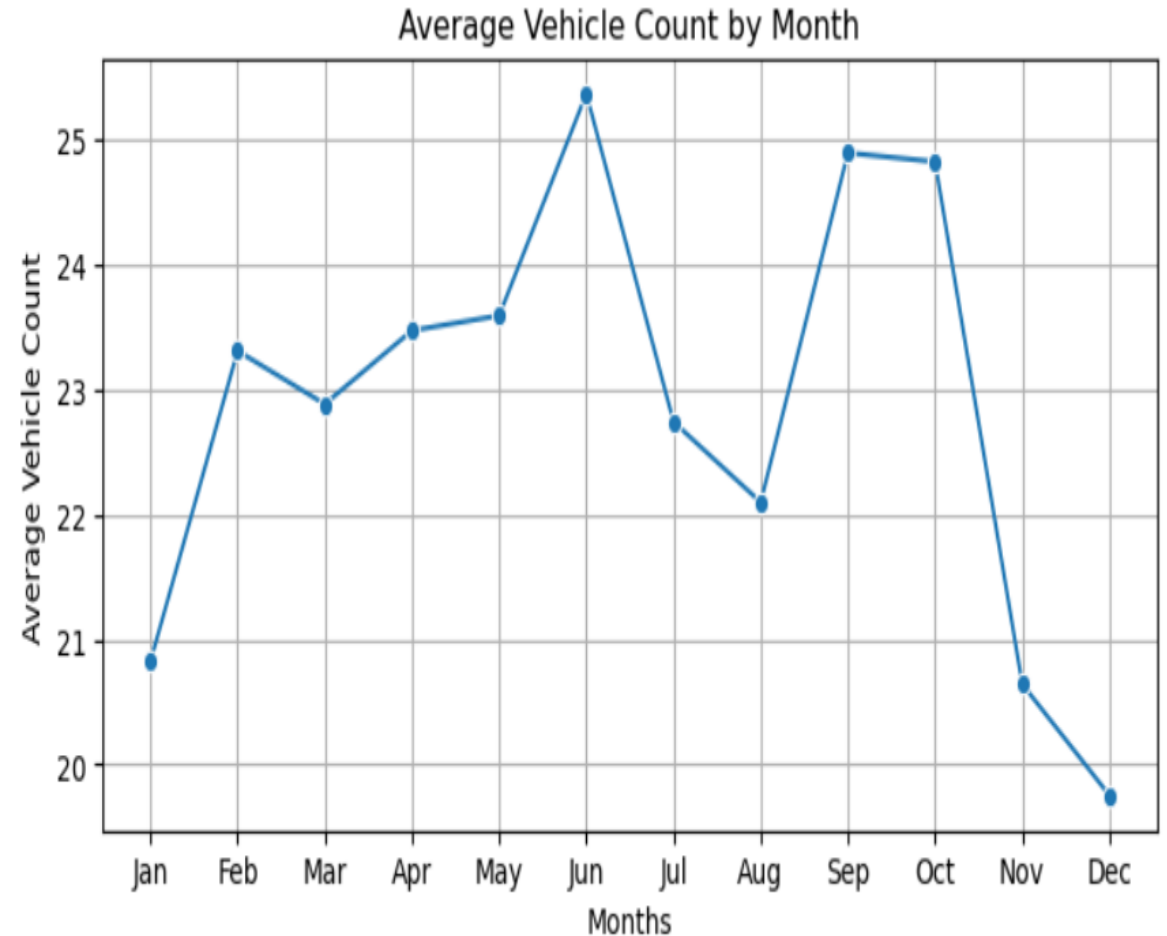
Low: The lowest average vehicle count occurs in December.

Seasonal Variation: The graph suggests a seasonal pattern in vehicle count, with higher counts during the summer months and lower counts during the winter months.

Possible Explanations:

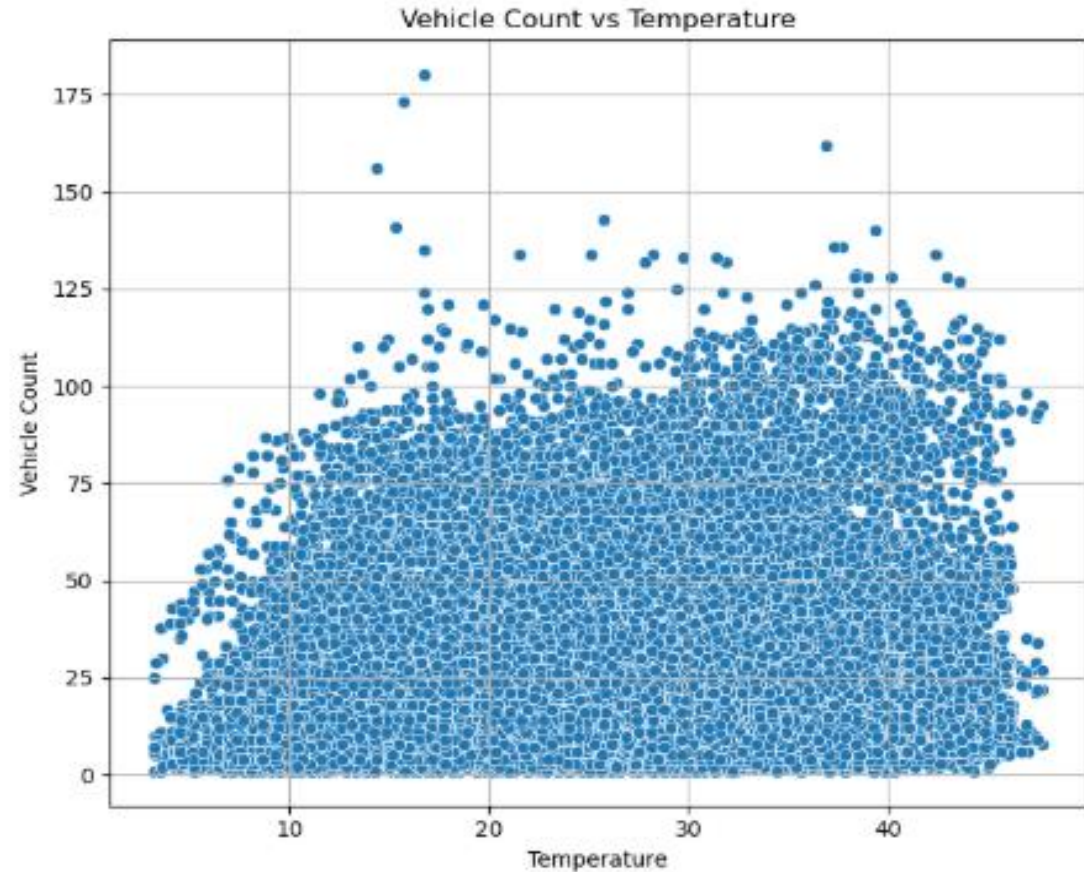
- Summer holidays and vacations:** More people travel during the summer months, resulting in a higher vehicle count.

- Winter weather:** In colder climates, people tend to drive less during the winter months due to snow and ice.



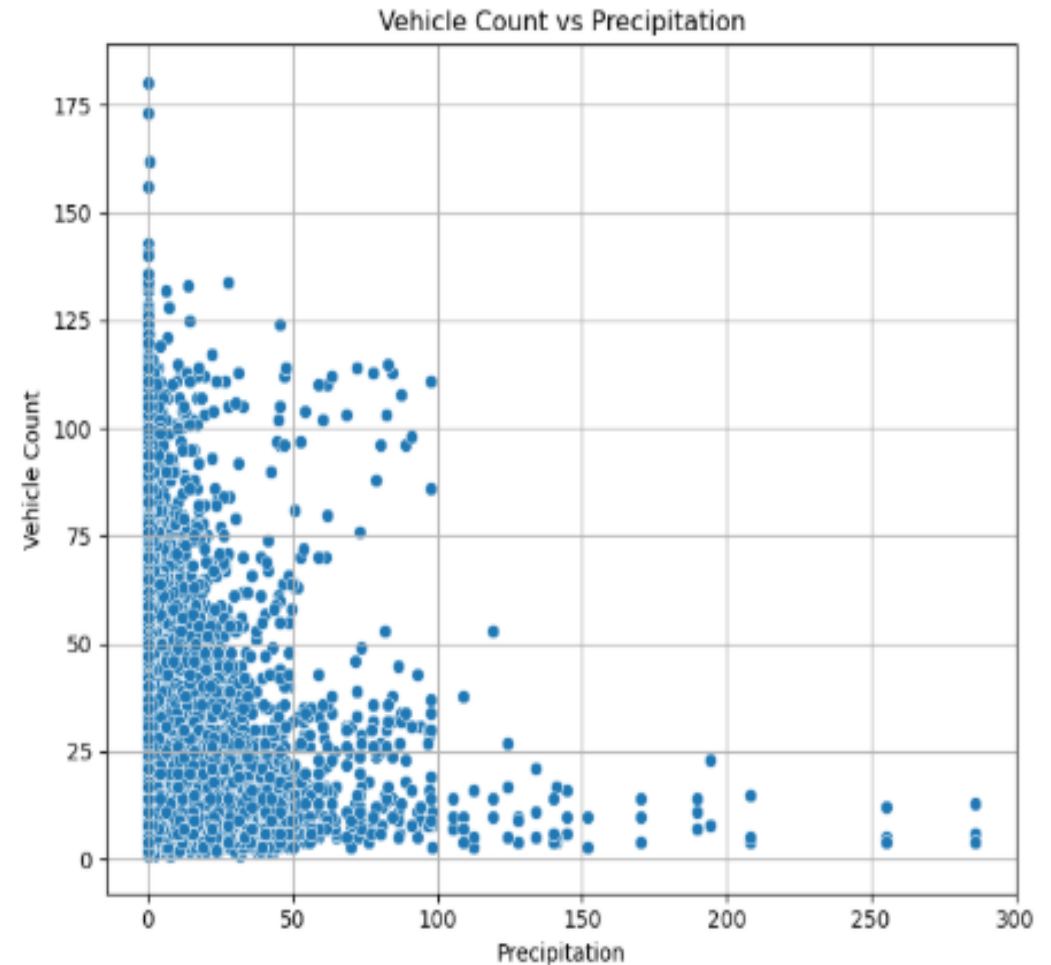
Temperature impact on traffic volumes

- There is a positive correlation between temperature and vehicle count.
- As the temperature increases, the number of vehicles on the road also tends to increase.



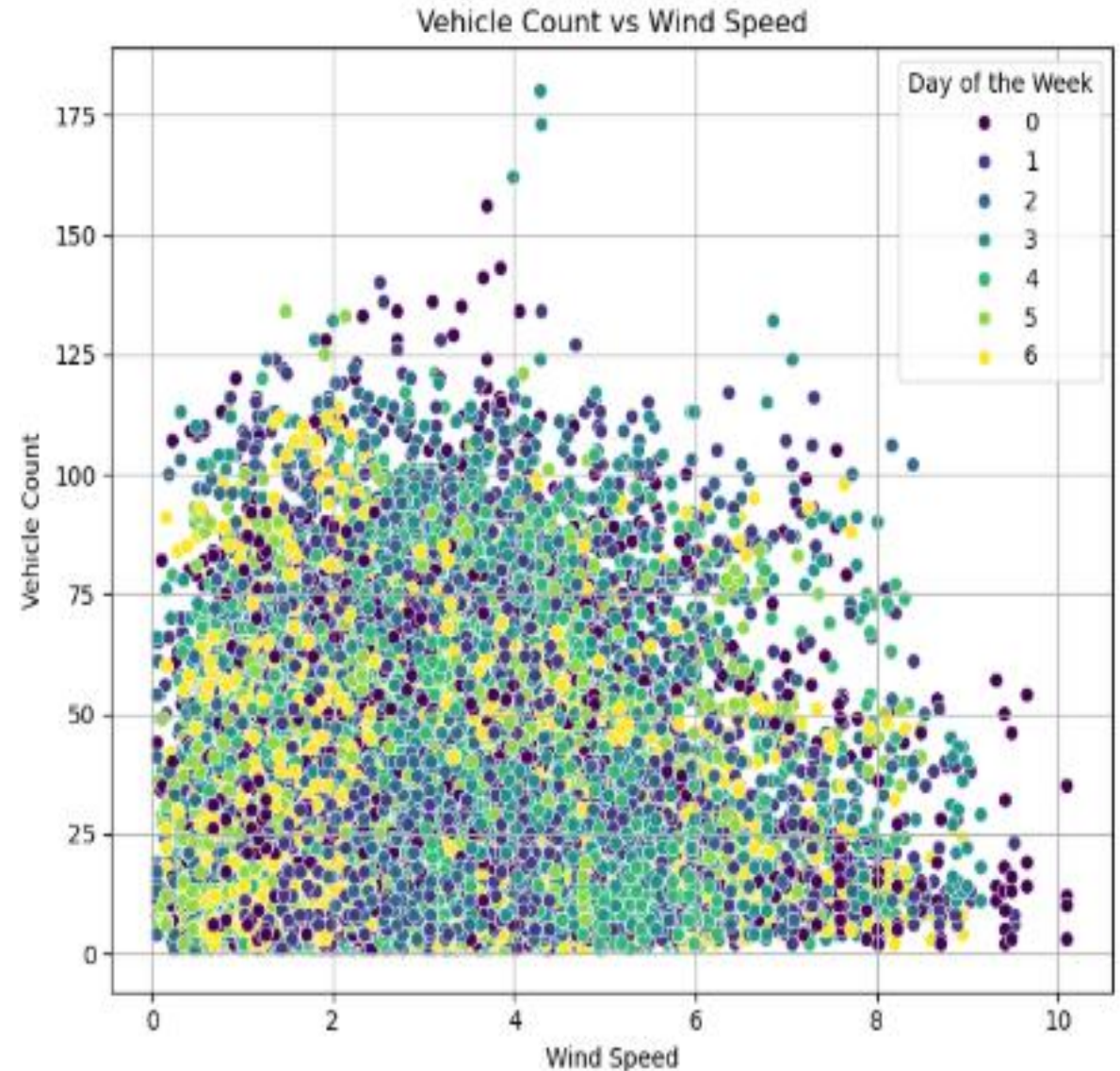
Precipitation impact on traffic volumes

This graph shows a negative correlation between vehicle count and precipitation. As precipitation increases, the vehicle count tends to decrease. This suggests people might be less likely to drive in heavy precipitation.



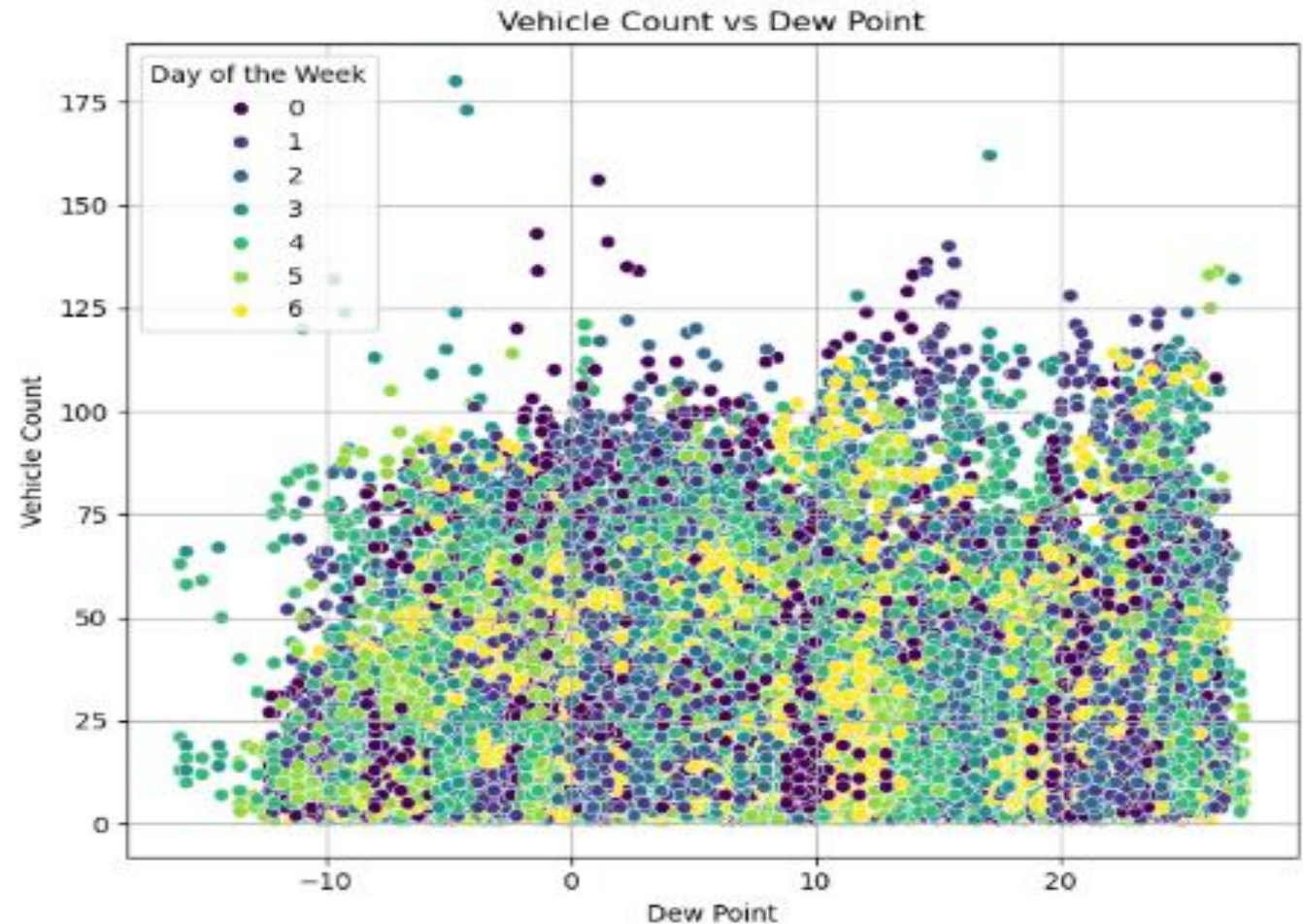
Wind effect on traffic volumes

- There is a negative correlation between wind speed and vehicle count. This means that as wind speed increases, the number of vehicles on the road tends to decrease.
- There is a high concentration of data points at lower wind speeds (less than 3 mph) with a wide range of vehicle counts. As wind speed increases, the range of vehicle counts narrows and the overall count decreases.



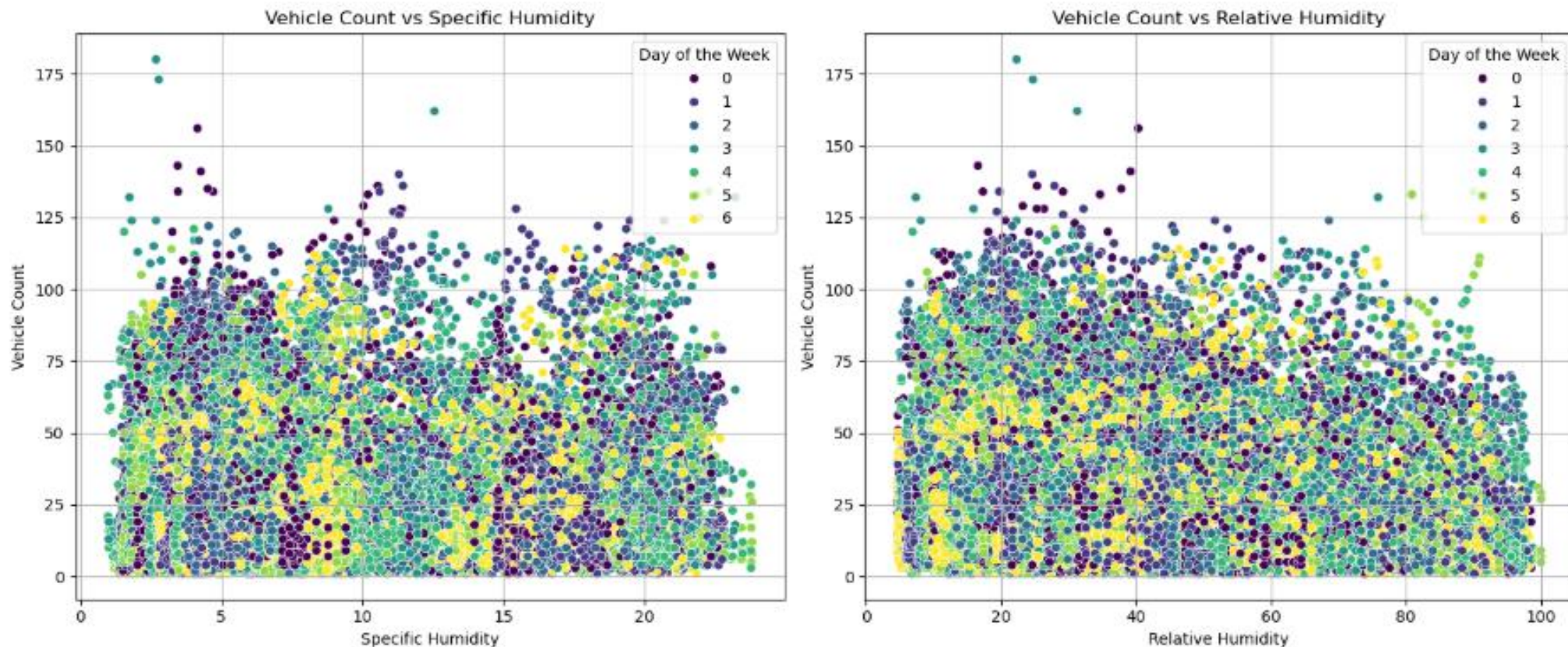
Dew Point analysis

- No clear correlation:** There doesn't appear to be a strong relationship between dew point and vehicle count.
- Even Distribution:** Points are fairly evenly distributed across the graph, suggesting dew point may not be a significant factor influencing vehicle count.



Humidity impact on traffic volumes

The graphs suggest that humidity (both specific and relative) has a limited influence on vehicle count. While there are subtle trends observed, the scattered data points indicate that other factors may play a more significant role in determining vehicle count.

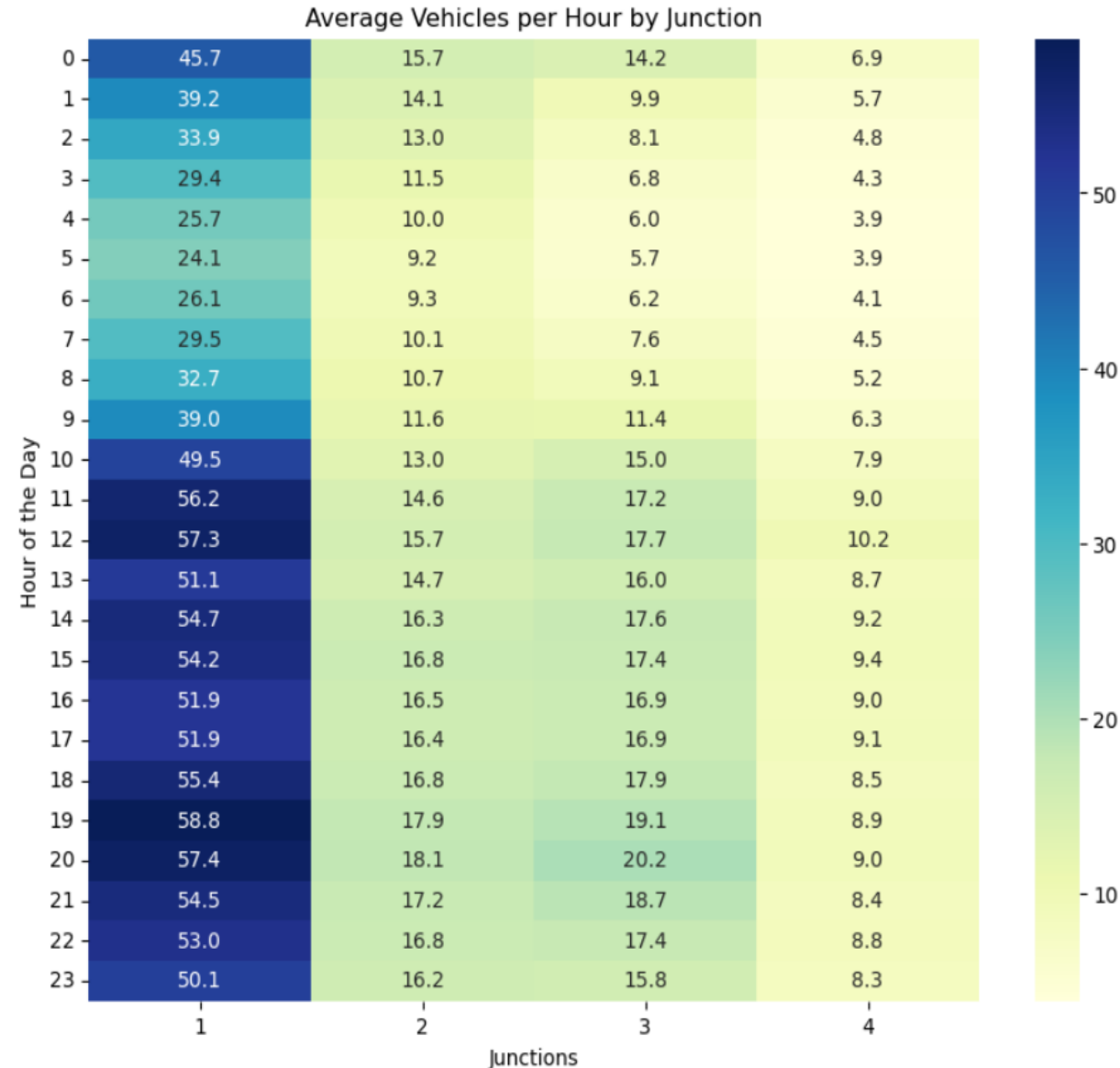


Peak hours analysis by Junctions

The number of vehicles passing through each junction generally increases as the hour of the day progresses, reaching a peak around 6 PM and then decreasing later in the evening.

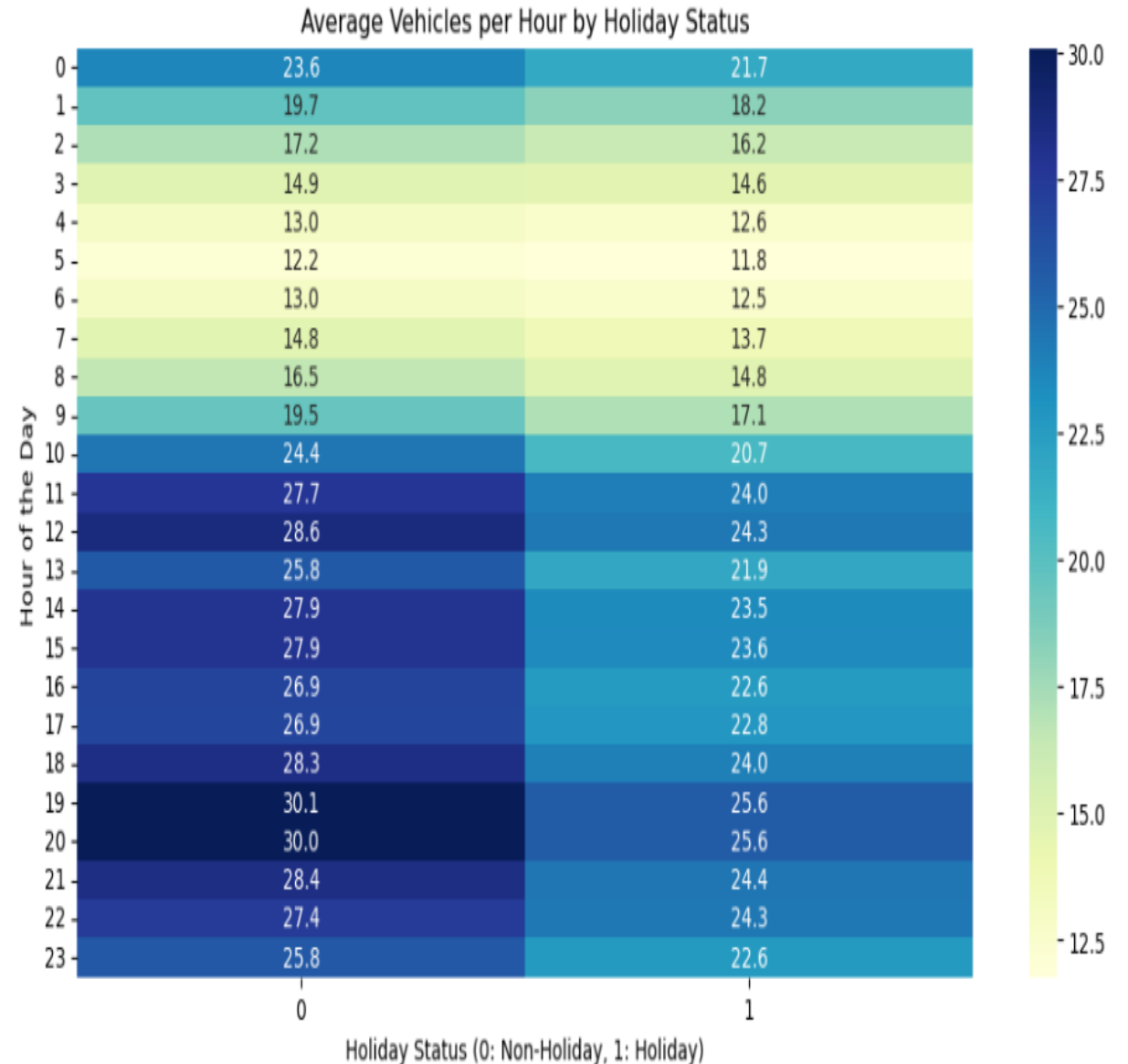
Junction Variations:

- Junction 1:** The highest traffic volume is consistently observed at Junction 1, particularly in the late afternoon and early evening hours.
- Junction 2:** Traffic volume at Junction 2 is significantly lower compared to Junction 1, showing a more gradual increase throughout the day.
- Junction 3:** Junction 3 shows a similar trend to Junction 2, with moderate traffic volume increasing steadily throughout the day.
- Junction 4:** Junction 4 exhibits the lowest traffic volume among all four junctions.



Holiday impact on traffic volume

- Traffic volume is generally higher during non-holiday hours compared to holiday hours.
- During non-holiday hours, peak traffic occurs between 18:00 and 21:00, with an average of 30 vehicles per hour.
- During holiday hours, peak traffic occurs between 11:00 and 15:00, with an average of around 24 vehicles per hour.
- Traffic volume is consistently lower during holiday hours, indicating a significant decrease in vehicle traffic during holiday periods.



Recommendations for traffic management and congestion mitigation during peak times.

1. Peak Traffic (Morning, Afternoon, Evening)

- Dynamic Pricing:** Implement surge pricing during peak hours (6 AM, 12 PM, 8 PM) to manage demand and incentivize drivers to operate during busy periods.
- Driver Incentives:** Offer bonuses or incentives for drivers to work during high-demand times.
- Ridesharing Options:** Encourage shared rides during peak hours to reduce the number of vehicles on the road.

2. Weekends vs. Weekdays

- Flexible Scheduling:** Promote flexible driver hours to address higher traffic on weekdays and lower traffic on weekends.
- Targeted Marketing:** Send notifications to riders on weekends offering discounts for off-peak travel.

3. Seasonal Trends (Summer & Winter)

- Adjust Driver Availability:** In winter, ensure sufficient drivers are available despite lower demand. In summer, plan for more drivers to meet increased demand.
- Vehicle Maintenance:** Ensure drivers maintain vehicles, especially in winter, to prevent breakdowns during harsh weather.

4. Weather Impacts (Temperature, Wind, Precipitation)

- Weather-Based Surge Pricing:** Increase surge pricing during bad weather (rain, snow) to balance supply and demand.
- Safety Alerts for Drivers:** Send weather alerts to drivers, advising caution during storms or high winds.

5. Junction Traffic

- Route Optimization:** Use real-time traffic data to guide drivers through less congested routes, especially around busy junctions.
- Avoid High-Traffic Areas:** Provide alternate routes for drivers near known high-traffic areas (like Junction 1 during evening peaks).

6. Holiday Traffic

- Adjust Surge Pricing:** Offer lower surge pricing during holidays (11 AM to 3 PM) to encourage riders to travel outside peak times.
- Incentivize Drivers:** Offer higher rates for drivers during non-holiday hours and peak holiday travel times.

7. General Measures

- Carpool Promotion:** Encourage riders to opt for carpooling to reduce congestion and costs.
- Optimize Dispatch System:** Use real-time traffic data and machine learning to dispatch drivers more efficiently during peak hours.

These strategies can help in managing peak traffic times more effectively, improve driver efficiency, and reduce congestion.